EFFICIENTNET-B3 BRAIN TUMOR CLASSIFICATION EXPERIMENT

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree
of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING



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October 2026

VIT BHOPAL UNIVERSITY, KOTHRIKALAN, SEHORE
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BONAFIDE CERTIFICATE

Certified that this project report titled "EfficientNet-B3 Brain Tumor Classification Experiment" is the bonafide work of "Hanish Singla (Register No: 25BCE10632), Abhash Pandey (Register No: 25BCE11412), Angelina Mishra (Register No: 25BCE10663), Bhoomi Mittal (Register No: 25BCE10391), Ankur Mishra (Register No: 25BCE11174)" who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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PROJECT GUIDE

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The Project Exhibition I Examination is held on _____

ACKNOWLEDGEMENT

First and foremost we would like to thank the Lord Almighty for His presence and immense blessings throughout the project work.

We wish to express our heartfelt gratitude to Dr. Vikas Panthi, Program Chair, School of Computer Science and Engineering, for his valuable support and encouragement in carrying out this work.

We would like to thank our internal guide, Dr. Vikas Panthi, for continually guiding and actively participating in our project and giving valuable suggestions to complete the project work.

We would like to thank all the technical and teaching staff of the School of Computing Science and Engineering, who extended directly or indirectly all support.

Last, but not least, we are deeply indebted to our parents who have been the greatest support while we worked day and night for the project to make it a success.

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
BN	Batch Normalization
BRISC	Brain tumor segmentation and classification dataset (BRISC2025)
CNN	Convolutional Neural Network
FC	Fully Connected (linear) layer
FN	False Negative
FP	False Positive
F1	F1-score (harmonic mean of precision and recall)
GAP	Global Average Pooling
GELU	Gaussian Error Linear Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
MPS	Metal Performance Shaders (Apple Silicon GPU backend)
MRI	Magnetic Resonance Imaging
PE1	Project Exhibition 1 (the proposed EfficientNet-B3 based model)
TP	True Positive

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ABSTRACT

Purpose: Brain tumor classification from magnetic resonance imaging (MRI) is a challenging image classification task because different tumor categories can look visually similar. This project investigates whether enlarging an EfficientNet backbone and modifying its classification head improves four-class brain tumor classification on the BRISC2025 dataset. The objective is to compare a modified EfficientNet-B3 model, called PE1, with an EfficientNet-B1 baseline under identical experimental conditions.

Methodology: PE1 uses an ImageNet-pretrained EfficientNet-B3 backbone followed by global average pooling, batch normalization, dropout, a 512-unit fully connected layer with GELU activation, a second dropout layer and a final four-class output layer. The classes are glioma, meningioma, no tumor and pituitary tumor. The 5,000-image training portion of BRISC2025 is divided into 4,000 training and 1,000 validation images, and the official 1,000-image test set is kept completely separate. Both models are trained and evaluated with the same protocol.

Findings: PE1 achieved an accuracy of 98.70% and a macro F1-score of 98.71% on the official test set, compared with 99.20% and 99.31% for the EfficientNet-B1 baseline. PE1 contains 11,488,300 parameters against 6,518,308 for the baseline (an increase of 76.25%), yet it made 13 test errors against 8. The larger architecture therefore did not improve classification performance under the evaluated conditions, which shows the importance of controlled architectural evaluation for brain tumor MRI classification.

Keywords: brain tumor; magnetic resonance imaging; deep learning; EfficientNet-B3; image classification; BRISC2025

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CHAPTER 1
PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

Brain tumor classification using magnetic resonance imaging (MRI) is an image classification task that involves distinguishing between tumor categories on the basis of their visual characteristics. Because different categories can appear similar in an image, the task is difficult even for experienced observers. Deep learning models, particularly convolutional neural networks (CNNs), learn visual features directly from image data and have therefore been widely explored for this purpose [3]–[7].

The performance of a classification model depends on its architecture and capacity, so any architectural modification has to be evaluated experimentally to find out whether the added capacity actually provides a benefit. This project follows that approach: it compares a compact EfficientNet-B1 baseline [1], [12] with a larger EfficientNet-B3 based model, named Project Exhibition 1 (PE1), on the BRISC2025 dataset [13]. The study concentrates on four categories: glioma, meningioma, no tumor and pituitary tumor. Both models solve the same four-class problem and are evaluated with the same protocol, so observed differences in performance can be attributed to the model architecture and not to a change in the task.

1.2 Motivation for the Work

Reading an MRI scan takes a considerable amount of time, depends on the skill of the radiologist and can vary between observers. The appearance of a tumor also changes with the scanning protocol and the imaging angle, and tumors of different types may have similar intensity and shape on a single two-dimensional slice. These difficulties motivate classification systems that can support clinical decision making.

A common assumption in deep learning is that a larger model gives better results. The baseline pipeline already reaches an accuracy of 0.9920 on the BRISC2025 test set with only 6,518,308 parameters [12], so the practical question is not whether the task can be solved, but whether a larger backbone with a deeper head can improve on this strong result and at what cost in parameters, training time and inference time. Answering this needs a controlled comparison in which only the model changes, which is the motivation for this project.

1.3 About the Project and the Techniques Used

The project is an experimental comparison between two models for four-class brain tumor MRI classification. The techniques used are summarised below.

- **Transfer learning:** both models start from weights pretrained on ImageNet [2] and are fine-tuned on MRI images.

- **EfficientNet backbones:** EfficientNet [8] scales network depth, width and input resolution together (compound scaling). B1 is used for the baseline and B3 for PE1, loaded from the timm library [14].
- **Modified classification head:** PE1 replaces the single linear layer of the baseline with batch normalization [9], dropout [10], a 512-unit fully connected layer and GELU activation [11], followed by a second dropout layer and a four-unit output layer.
- **Training:** the Adam optimizer [15] with cross-entropy loss, a ReduceLROnPlateau learning-rate schedule and checkpoint selection on the validation set, implemented in PyTorch [17].
- **Evaluation:** accuracy, macro and weighted precision, recall and F1-score, test loss, parameter count, training duration and inference time per image.

Table 1.1: Project at a glance

Item	Description
Task	Four-class brain tumor classification from MRI images (glioma, meningioma, no tumor, pituitary)
Dataset	BRISC2025: 5,000 training images and 1,000 official test images [13]
Baseline	EfficientNet-B1 with dropout and a single linear layer (6,518,308 parameters)
Proposed model (PE1)	EfficientNet-B3 with a modified classification head (11,488,300 parameters)
Input	Three-channel image, 224 × 224 pixels
Evaluation	Official 1,000-image test set, best validation checkpoint
Repository	github.com/Belugabilli/BrainTumor-PE-1 [16]

1.4 Problem Statement

Given a two-dimensional brain MRI image, assign it to one of four classes: glioma, meningioma, no tumor or pituitary tumor. The specific problem addressed is to determine whether replacing an EfficientNet-B1 backbone with EfficientNet-B3 and a deeper, more regularised classification head improves classification performance on the BRISC2025 dataset under identical data splits, preprocessing, optimizer settings and evaluation method, and to measure the cost of any such change in parameters, training time and inference time.

1.5 Objective of the Work

- To develop an EfficientNet-B3 based model with a modified classification head for four-class brain tumor MRI classification.
- To compare the proposed PE1 model with an EfficientNet-B1 baseline under the same experimental conditions.
- To evaluate both architectures with standard classification metrics, including per-class results.
- To analyse parameter count, training duration and inference time alongside classification performance, in order to assess the effect of the architectural modification.

1.6 Organization of the Project

The report is organised in seven chapters. Chapter 1 describes the project and its objectives. Chapter 2 reviews related work. Chapter 3 lists the requirements of the experiment. Chapter 4 explains the design methodology. Chapter 5 presents the technical implementation, testing and performance analysis. Chapter 6 discusses the project outcome and its applicability. Chapter 7 gives the conclusions, limitations and recommendations. References and appendices follow.

1.7 Summary

This chapter introduced MRI-based brain tumor classification, explained why a controlled comparison between EfficientNet-B1 and EfficientNet-B3 is needed, and stated the problem and objectives of the project.

CHAPTER 2
RELATED WORK INVESTIGATION

2.1 Introduction

This chapter reviews the work on which the project is built: the use of CNNs and transfer learning for brain tumor MRI classification, the EfficientNet family of backbones, the components of the modified classification head, and the baseline pipeline on the BRISC2025 dataset. It ends by stating the gap that the project addresses.

2.2 Core Area of the Project

The core area of the project is medical image classification with deep learning, specifically the classification of brain MRI images into tumor categories. Gliomas, meningiomas and pituitary tumors make up most of the primary brain tumors seen in imaging. They differ in where they start, how they grow and how they are treated, so distinguishing between them is essential for treatment planning. MRI is the preferred modality because it gives excellent soft-tissue contrast without ionising radiation.

CNNs learn feature maps directly from images, which removes the need for hand-made texture or shape descriptors. Medical datasets are however much smaller than collections of natural images because expert labels are costly, and a network trained from scratch on such data overfits easily. Transfer learning addresses this by initialising the network with ImageNet weights [2] and then adapting it, either by training only the new classification layers or by fine-tuning part or all of the backbone.

2.3 Existing Approaches and Methods

2.3.1 Fine-tuned pretrained CNNs for MRI classification

A number of studies apply pretrained CNNs to brain tumor MRI. Swati et al. [3] used transfer learning for content-based retrieval of brain tumor MR images. Filatov and Yar [4] studied brain tumor diagnosis and classification with pretrained CNNs. Gómez-Guzmán et al. [5] classified brain tumors on MRI with CNNs, Poudyal et al. [6] presented a comparative analysis of transfer learning for MRI-based classification, and Chen et al. [7] applied deep learning to brain tumor detection and classification. Fine-tuned ImageNet backbones have become the usual reference approach for this task.

2.3.2 EfficientNet backbones and compound scaling

EfficientNet [8] studies how a CNN should grow when more computing power is available. Earlier methods increased only one dimension, such as depth, width or input resolution, which gave diminishing returns. Tan and Le proposed compound scaling, in which depth, width and resolution are scaled together with fixed coefficients starting from a B0 network found by neural architecture search. This produced the family B1 to B7, where each larger

model is designed for a larger input. EfficientNet-B1 is designed for images of about 240 × 240 pixels and B3 for about 300 × 300 pixels, and B3 has more layers and wider connections and therefore many more parameters.

2.3.3 EfficientNet-B1 baseline on BRISC2025

Alkharaan et al. [1] address brain tumor classification and segmentation using EfficientNet and U-Net++ models, and their study, together with the BRISC dataset [13], forms the starting point of this project. The baseline pipeline [12] uses an EfficientNet-B1 model trained on BRISC2025. On the 1,000-image test set it achieves an accuracy of 0.9920 and a macro F1-score of 0.9931 with 6,518,308 parameters. Training for 50 epochs takes 8,674.3 seconds (about 144.6 minutes) and inference takes 1.66 milliseconds per image on an Apple Silicon device using MPS.

2.3.4 Classification head design and regularisation

In transfer learning pipelines the pretrained backbone usually ends with average pooling, after which a classification head turns the feature vector into class scores. The common choice is a single linear layer, although a deeper head is sometimes used to add capacity and regularisation. Batch normalization [9] standardises activations and makes training more stable, dropout [10] randomly switches off units during training to reduce overfitting, and the Gaussian error linear unit (GELU) [11] provides a smooth non-linearity that is popular in recent models.

2.4 Pros and Cons of the Stated Approaches

Table 2.1: Pros and cons of the approaches reviewed

Approach	Advantages	Limitations
Fine-tuned pretrained CNNs	Needs less labelled data, converges faster, reuses general ImageNet filters	Domain gap between natural images and MRI scans; backbone cost
Compound-scaled EfficientNet backbones	Good accuracy-to-parameter trade-off; family of sizes from B0 to B7	Scaling was tuned on ImageNet, so the gain on a small specialised dataset is uncertain; larger variants expect larger inputs
Deeper regularised heads (BN, dropout, GELU)	Extra capacity and regularisation at the point of task adaptation	Adds parameters; benefit for generalisation on a moderate MRI dataset has to be tested

2.5 Issues and Observations from the Investigation

The baseline pipeline shows that a small EfficientNet-B1 model already classifies BRISC2025 images with high accuracy. It does not show whether a bigger backbone together with a deeper, more regularised head would improve performance, nor what that would cost in parameters, training time and inference speed. Compound scaling was tuned

on ImageNet, so it is not clear that a larger model helps on a small specialised dataset, particularly when images are resized below the resolution the larger model was designed for.

Answering these questions requires a comparison where only the model changes and the data split, preprocessing, optimizer settings and evaluation method stay unchanged. This is the gap addressed by PE1. Because the backbone and the head are changed together, the comparison reflects the combined effect of both changes and not the contribution of each part individually.

2.6 Summary

Transfer learning with EfficientNet backbones is an established approach to brain tumor MRI classification, and a compact B1 baseline already performs strongly on BRISC2025. The open question is whether a larger backbone with a deeper head offers any benefit, and this is the question the project investigates.

CHAPTER 3
REQUIREMENT ARTIFACTS

3.1 Introduction

This chapter lists the hardware, software, data, functional, performance and security requirements of the experiment. As the project is a deep learning experiment and not an end-user application, the requirements concern the experimental pipeline.

3.2 Hardware and Software Requirements

Table 3.1: Hardware and software used in the experiment

Component	Details
Compute device	Apple Silicon device; the MPS backend was used for training and for measuring inference time
Programming language	Python
Deep learning framework	PyTorch [17]
Model library	timm (PyTorch Image Models) [14] for the pretrained EfficientNet backbones
Metrics	scikit-learn [18] for classification metrics
Version control	GitHub repository [16]

3.3 Specific Project Requirements

3.3.1 Data requirement

The experiment uses the BRISC2025 brain MRI classification dataset [13]. The training folder contains 5,000 images and a separate official test folder contains 1,000 images, giving 6,000 images in total across four classes (Table 3.2). Labels are encoded as glioma = 0, meningioma = 1, no_tumor = 2 and pituitary = 3. The 5,000-image training folder is split 80:20 with a fixed random seed of 42 into 4,000 training and 1,000 validation images. The official test images are never used for training, validation or checkpoint selection.

Table 3.2: Class distribution of the BRISC2025 dataset used in the project

Class	Training folder	Official test	Total
Glioma	1,147	254	1,401
Meningioma	1,329	306	1,635
No tumor	1,067	140	1,207
Pituitary tumor	1,457	300	1,757
Total	**5,000**	**1,000**	**6,000**

3.3.2 Functional requirements

- Accept a three-channel MRI image at a fixed resolution of 224 × 224 pixels.
- Produce a prediction for exactly one of four labels: glioma, meningioma, no tumor or pituitary tumor.

- Train both the baseline and PE1 under the same protocol and keep the checkpoint with the best validation performance.
- Evaluate the selected checkpoint on the untouched official test set and record accuracy, macro and weighted precision, recall and F1-score, test loss, parameter count, training duration and inference time per image.

Tumor segmentation and pixel-level boundary detection are outside the scope of the project.

3.3.3 Performance and security requirements

Performance. The reference for comparison is the baseline accuracy of 0.9920 and macro F1-score of 0.9931. The comparison must be fair: both models use the same data split, input size, optimizer, loss function, batch size, number of epochs and random seed. Computational cost is measured through parameter count, 50-epoch training time and per-image inference time.

Reproducibility and data integrity. A fixed seed (42) is used throughout, and the official test set must remain isolated from training and model selection so that no information leaks into the reported results.

Security and privacy. No additional patient data is collected or processed; the experiment uses only the BRISC2025 dataset. Any clinical deployment of such a model would need to meet data-protection and regulatory requirements, which are outside the scope of this project.

3.3.4 Look and feel requirements

The project has no graphical user interface. Results are produced as experiment logs and metric reports, and presented in this report as tables and charts.

3.4 Summary

The experiment needs a single Apple Silicon machine with a PyTorch and timm environment, the BRISC2025 dataset with an isolated official test set, and a pipeline that records both classification metrics and computational cost for two models under identical conditions.

CHAPTER 4

DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

The project follows a controlled-comparison methodology. The goal is to examine whether a larger EfficientNet backbone, combined with a redesigned classification head, improves four-class brain tumor classification from MRI images. The proposed PE1 model is evaluated against the EfficientNet-B1 baseline. Both models address the same task and use the same dataset partitions and principal training settings, so the effect of the model configuration can be examined without changing the target classes or the test set.

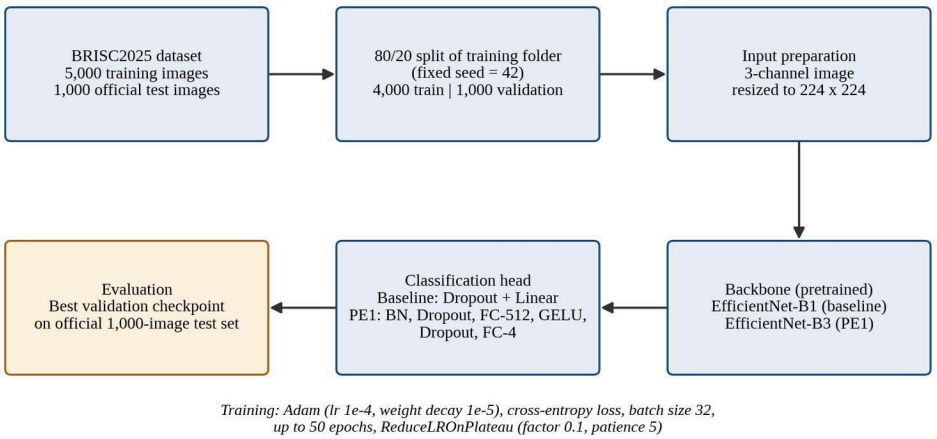


Figure 4.1: Experimental pipeline used for both models

4.2 Functional Modules Design and Analysis

The pipeline is divided into the functional modules listed in Table 4.1. Only the model definition module differs between the baseline and PE1.

Table 4.1: Functional modules of the experimental pipeline

Module	Function	Differs between models?
Data module	Reads the BRISC2025 folders and performs the 80/20 training and validation split with seed 42	No
Input preparation	Provides three-channel 224 × 224 images and the label mapping (0 to 3)	No
Model definition	Builds the baseline (B1 + dropout + linear) or PE1 (B3 + modified head)	Yes
Training module	Adam optimizer, cross-entropy loss, ReduceLROnPlateau, up to 50 epochs, best-validation checkpoint	No
Evaluation module	Computes accuracy, precision, recall, F1-score, test loss, parameters, training time and inference time on the official test set	No

4.3 Software Architectural Design

Figure 4.2 compares the two network architectures. The baseline passes the 1,280-dimensional pooled feature vector of EfficientNet-B1 through a dropout layer (p = 0.30) and a linear layer to four class scores. PE1 replaces the backbone with EfficientNet-B3, which produces a 1,536-dimensional pooled vector, and replaces the single layer by the head described in Section 4.4.

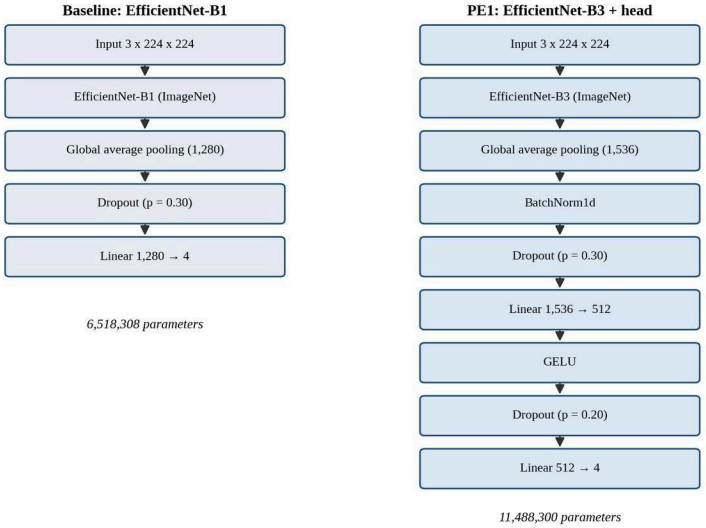


Figure 4.2: Architecture of the EfficientNet-B1 baseline and the proposed PE1 model

Table 4.2: Main architectural differences between the baseline and PE1

Feature	EfficientNet-B1 baseline	PE1 (EfficientNet-B3)
Backbone	EfficientNet-B1 (pretrained)	EfficientNet-B3 (pretrained)
Pooled feature size	1,280	1,536
Classification head	Dropout (0.30) → Linear (1,280, 4)	BatchNorm → Dropout (0.30) → Linear (1,536, 512) → GELU → Dropout (0.20) → Linear (512, 4)
Total parameters	6,518,308	11,488,300
Trainable parameters	6,518,308	11,488,300
Input dimensions	3 × 224 × 224	3 × 224 × 224

4.4 Subsystem Services

The main subsystem of PE1 is the classification head. After global average pooling, the 1,536-dimensional feature vector passes through batch normalization to keep optimisation stable, a dropout layer with rate 0.30 to reduce overfitting, a fully connected layer that reduces 1,536 units to 512, a GELU activation that provides a smooth non-linearity, a second dropout layer with rate 0.20, and a final linear layer that outputs four scores for glioma, meningioma, no tumor and pituitary tumor.

The head contains 3,072 batch normalization parameters, 786,944 parameters in the 1,536 → 512 layer and 2,052 parameters in the output layer, a total of 792,068. The remaining

10,696,232 parameters belong to the EfficientNet-B3 backbone, giving the reported total of 11,488,300. The baseline head has 5,124 parameters, so its backbone has 6,513,184.

4.5 User Interface Design

Not applicable. The project is a command-line experiment with no user-facing interface; results are written to logs and metric reports.

4.6 Novelty of the Design

The contribution of the project is not a new network family but a controlled and fully documented comparison. The novelty lies in (i) a modified EfficientNet-B3 classification head with batch normalization, two dropout stages and a 512-unit GELU layer, (ii) the evaluation of this head and the larger backbone against a strong B1 baseline under exactly the same split, seed, optimizer, loss and test set, and (iii) the reporting of computational cost (parameters, training time and inference time) together with classification metrics, which shows that a larger model does not automatically give better results.

4.7 Summary

The design isolates the model architecture as the only changing factor. PE1 combines an EfficientNet-B3 backbone with a deeper regularised head, and the baseline uses EfficientNet-B1 with a single linear layer; everything else in the pipeline is identical.

CHAPTER 5
TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Outline

This chapter describes the implementation of PE1, the training configuration, the validation of the setup, and the performance analysis of both models on the official test set.

5.2 Technical Coding and Code Solutions

Listing 5.1 shows the PE1 model definition in PyTorch using the timm library, and Listing 5.2 the training loop. The listings are illustrative and consistent with the architecture and settings documented for the project; the complete code is available in the project repository [16].

Listing 5.1: PE1 model definition

```
import torch.nn as nn, timm

class PE1(nn.Module):
    def _init_(self, num_classes=4): super().
        init_()
        # ImageNet-pretrained EfficientNet-B3, pooled 1,536-d output
        self.backbone = timm.create_model(
            "efficientnet_b3", pretrained=True, num_classes=0)
        self.head = nn.Sequential(
            nn.BatchNorm1d(1536), nn.Dropout(0.30), nn.Linear(1536,
            512), nn.GELU(), nn.Dropout(0.20), nn.Linear(512,
            num_classes))

    def forward(self, x):              # x: (N, 3, 224, 224)
        return self.head(self.backbone(x))
```

Listing 5.2: Training configuration and loop

```
torch.manual_seed(42)
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4, weight_decay=1e-5) criterion
= nn.CrossEntropyLoss()
scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau( optimizer,
    mode="min", factor=0.1, patience=5)
best = float("inf")
for epoch in range(50):       # maximum 50 epochs, batch size 32
    train_one_epoch(model, train_loader, optimizer, criterion) val_loss
    = evaluate(model, val_loader, criterion) scheduler.step(val_loss)
    if val_loss < best:       # keep best validation checkpoint best =
        val_loss
        torch.save(model.state_dict(), "best.pt")
```

5.3 Working Layout of Forms

Not applicable. The experiment is executed as scripts and has no input forms or screens.

5.4 Prototype Submission

The prototype is the PE1 experiment repository, which contains the code, configuration and experiment summary: github.com/Belugabilli/BrainTumor-PE-1 [16]. The EfficientNet-B1 baseline is available at github.com/Belugabilli/BrainTumor-Reproduction [12].

5.5 Test and Validation

Both models were trained with the settings in Table 5.1. The training folder was split into 4,000 training and 1,000 validation images, and the checkpoint with the best validation performance was selected for evaluation; for PE1 this checkpoint was obtained at epoch 13 of a maximum of 50. The selected checkpoint was then evaluated once on the official 1,000-image test set, which was never used in training, validation or model selection. The evaluation measures (accuracy, precision, recall, F1-score with macro and weighted averages, test loss, parameter count, training time and inference time) are defined in Appendix B.

Table 5.1: Training settings used for both models

Setting	Value
Input resolution	224 × 224 pixels, three channels
Training / validation / official test	4,000 / 1,000 / 1,000 images
Batch size / maximum epochs	32 / 50
Optimizer	Adam [15]
Learning rate / weight decay	0.0001 (1×10^{-4}) / 0.00001 (1×10^{-5})
Loss function	Cross-entropy loss
Learning-rate scheduler	ReduceLROnPlateau; mode = min; factor = 0.1; patience = 5
Random seed	42

5.6 Performance Analysis (Graphs and Charts)

5.6.1 Overall test-set performance

Both models were evaluated on the same official test set of 1,000 MRI images. The EfficientNet-B1 baseline reached 99.20% accuracy and PE1 reached 98.70%, so the proposed B3 model did not improve overall accuracy. The difference corresponds to five images in favour of the baseline (8 errors against 13). The aggregate results are given in Table 5.2 and Figures 5.1 and 5.2.

Table 5.2: Aggregate test-set performance and computational comparison

Metric	B1 baseline	PE1 (B3 + head)	PE1 minus baseline
Accuracy	0.9920	0.9870	−0.0050
Macro precision	0.9930	0.9853	−0.0077
Macro recall	0.9932	0.9890	−0.0042
Macro F1-score	0.9931	0.9871	−0.0060
Weighted F1-score	0.9920	0.9870	−0.0050
Test loss	0.0236	0.0562	+0.0326
Total parameters	6,518,308	11,488,300	+4,969,992 (+76.25%)
Training time, 50 epochs	8,674.3 s (~144.6 min)	4,337.8 s (~72.3 min)	−4,336.5 s (−50.0%)
Inference time per image	1.66 ms (MPS)	1.63 ms (MPS)	−0.03 ms (−1.8%)

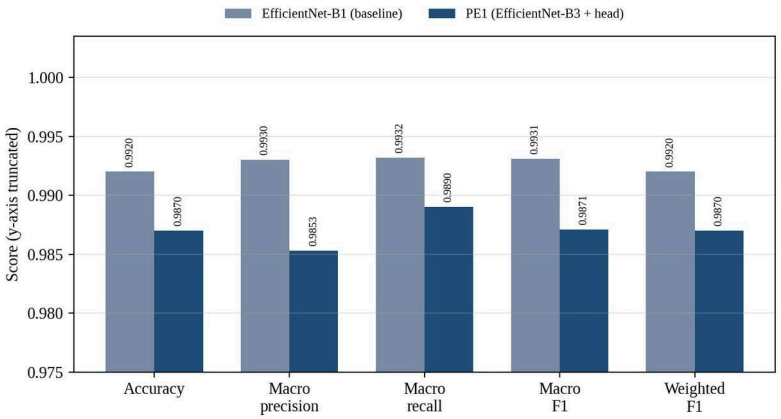


Figure 5.1: Aggregate test-set metrics of the baseline and PE1 (y-axis truncated)

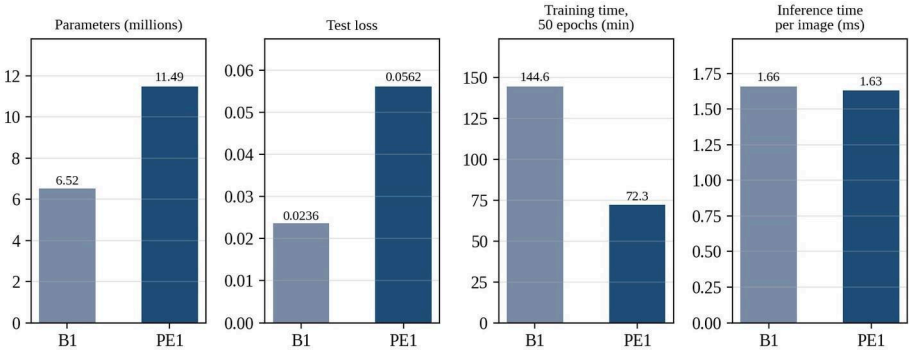


Figure 5.2: Parameter count, test loss, training time and inference time of the baseline and PE1

PE1 has 76.25% more parameters than the baseline, but the measured inference times are very close (1.63 ms against 1.66 ms per image). PE1 also has a higher test loss (0.0562 against 0.0236). The shorter 50-epoch training duration logged for PE1 describes the recorded experiment environment; a larger network would normally be expected to train more slowly, so this difference is most probably due to differences in the runtime environment between the two runs and should not be read as an architectural advantage.

5.6.2 Class-wise performance of PE1

Table 5.3 and Figure 5.3 show the per-class results of PE1. Glioma has the highest F1-score (0.9921), followed by pituitary tumor (0.9916), no tumor (0.9859) and meningioma (0.9787). The no-tumor class has a recall of 1.0000, meaning all 140 no-tumor images were identified correctly, but its precision of 0.9722 is the lowest of the four classes. Meningioma has the lowest recall (0.9771) and F1-score.

Table 5.3: Per-class metrics of PE1 on the official test set

Class	Precision	Recall	F1-score	Support
Glioma	0.9921	0.9921	0.9921	254
Meningioma	0.9803	0.9771	0.9787	306
No tumor	0.9722	1.0000	0.9859	140
Pituitary tumor	0.9966	0.9867	0.9916	300
Overall	**0.9871**	**0.9870**	**0.9870**	**1,000**

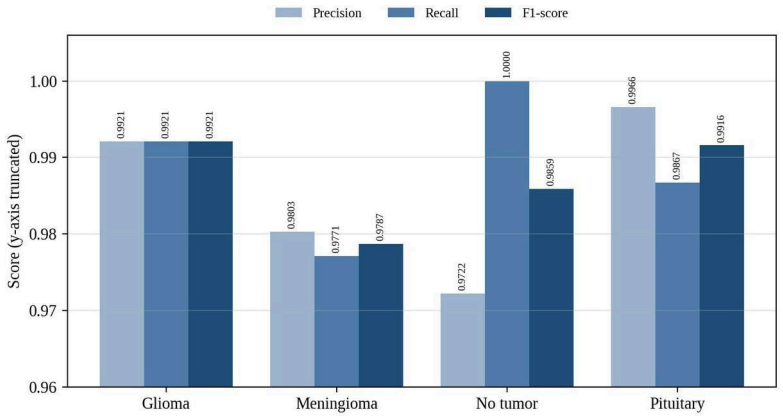


Figure 5.3: Per-class precision, recall and F1-score of PE1 (y-axis truncated)

Combining the recall and precision values with the class supports gives the approximate error counts in Table 5.4. They are derived values (recall × support for missed images and the precision relation for false alarms) and add up to the 13 test errors of PE1; the class-by-class confusion matrix is not reproduced here.

Table 5.4: Approximate error counts of PE1 derived from the per-class metrics

Class	Support	Correct	Missed	False alarms
Glioma	254	252	2	2
Meningioma	306	299	7	6
No tumor	140	140	0	4
Pituitary tumor	300	296	4	1
Total	**1,000**	**987**	**13**	**13**

5.7 Summary

PE1 was implemented as an EfficientNet-B3 backbone with a five-layer classification head and trained under the same protocol as the baseline. On the official test set the baseline was better on every aggregate classification metric, while PE1 required 76.25% more parameters and showed essentially the same inference time.

CHAPTER 6

PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This chapter summarises what was implemented, the significant outcomes of the experiment and how the findings can be applied.

6.2 Key Implementations Outlines of the System

- An EfficientNet-B3 based classifier (PE1) with a modified head of batch normalization, dropout, a 512-unit fully connected layer, GELU and a second dropout layer, containing 11,488,300 trainable parameters.
- A controlled training and evaluation pipeline shared with the EfficientNet-B1 baseline: same split (4,000 / 1,000 / 1,000), seed 42, Adam optimizer, cross-entropy loss and ReduceLROnPlateau scheduler.
- A complete evaluation on the isolated official test set, covering aggregate and per-class metrics, test loss, parameters, training time and inference time.

6.3 Significant Project Outcomes

- PE1 reached 98.70% accuracy and a macro F1-score of 0.9871, while the baseline reached 99.20% and 0.9931. The baseline was better on all aggregate classification metrics.
- PE1 made 13 errors on 1,000 test images against 8 for the baseline, even though it has 76.25% more parameters.
- Inference time was almost identical (1.63 ms against 1.66 ms per image), so the larger model gave neither a significant slowdown nor a speed advantage.
- Within PE1, meningioma was the weakest class (F1 0.9787), and no tumor had perfect recall but the lowest precision (0.9722), meaning about four tumor images were assigned to the no-tumor class. This is the more serious kind of error in practice because it could delay a diagnosis; since the class has only 140 test images, a few errors change its metrics noticeably.
- The key finding is that, on this dataset and under these conditions, model size is not the limiting factor for classification performance.

6.4 Project Applicability on Real-world Applications

- **Model selection:** the results support choosing a compact backbone such as EfficientNet-B1 for brain tumor MRI classification when a larger model gives no accuracy benefit; B1 has about 43% fewer parameters than PE1.
- **Decision support research:** the pipeline can serve as a basis for computer-aided decision-support prototypes for MRI triage, subject to clinical validation that is outside the scope of this project.

- **Experimental method:** the controlled-comparison protocol can be reused to evaluate other architectural changes in medical image classification, and to avoid assuming that added capacity helps.
- **Resource-limited settings:** a smaller model with equal or better accuracy is easier to train and deploy on limited hardware.

6.5 Inference

The experiment does not support the claim that a larger EfficientNet backbone and a deeper classification head automatically improve results on this dataset. The compact EfficientNet-B1 baseline remains the more appropriate choice for the evaluated task, because it achieves higher accuracy with fewer parameters.

CHAPTER 7
CONCLUSIONS AND RECOMMENDATION

7.1 Outline

This chapter presents the conclusions of the project, the limitations of the study and recommendations for future work.

7.2 Limitations and Constraints of the System

- **Single run:** all results come from one run with one fixed seed. A difference of 0.5 percentage points (five images out of 1,000) cannot be regarded as statistically significant without repeated runs, with mean and standard deviation reported.
- **Early convergence:** the best PE1 checkpoint was obtained at epoch 13 of 50, which suggests that the additional capacity of B3 is not used on a dataset of this size.
- **Input resolution:** images were resized to 224 × 224, which is closer to the resolution B1 is designed for than to B3's native resolution, and this may have limited B3.
- **Combined change:** the backbone and the head were changed together, so the individual contribution of each cannot be separated.
- **Timing comparison:** the training durations come from separate runs in a possibly different runtime environment and are not a reliable measure of efficiency.
- **Scope:** the study covers classification only. Segmentation and explainability (for example Grad-CAM [19]) were not assessed, and since the dataset is a single curated source, generalisation to other scanners and institutions has not been tested.

7.3 Future Enhancements

- Repeat the experiments with several random seeds, report mean and standard deviation, and apply a statistical test for comparing classifiers [20].
- Evaluate EfficientNet-B3 at its native input resolution (about 300 × 300 pixels).
- Run an ablation in which the backbone and the head are changed separately.
- Use class-weighted losses or targeted augmentation for the meningioma and no-tumor classes.
- Validate on external, multi-institution data to test generalisation.
- Integrate the classifier with a segmentation stage and add Grad-CAM explainability [19] to move towards a complete clinical workflow.

7.4 Inference

Under the same training and evaluation procedure, EfficientNet-B3 with an enhanced classification head did not outperform the EfficientNet-B1 baseline. The baseline obtained 99.20% accuracy with 6.52 million parameters, whereas PE1 reached 98.70% accuracy with

11.49 million parameters. EfficientNet-B1 therefore offers a good compromise between accuracy and compactness, and the additional parameters of B3 did not lead to better predictions on this dataset.

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APPENDIX A: EXPERIMENT CONFIGURATION AND REPOSITORIES

Item	Details
Project repository (PE1)	https://github.com/Belugabilli/BrainTumor-PE-1
Baseline repository (EfficientNet-B1)	https://github.com/Belugabilli/BrainTumor-Reproduction
Dataset	BRISC2025 (5,000 training and 1,000 test images, four classes)
Label encoding	glioma = 0, meningioma = 1, no_tumor = 2, pituitary = 3
Input	3 × 224 × 224
PE1 head	BatchNorm1d(1536) → Dropout(0.30) → Linear(1536, 512) → GELU → Dropout(0.20) → Linear(512, 4)
Optimizer / loss	Adam (lr 1e-4, weight decay 1e-5) / cross-entropy
Scheduler	ReduceLROnPlateau (mode min, factor 0.1, patience 5)
Batch size / epochs / seed	32 / maximum 50 / 42
Selected PE1 checkpoint	Epoch 13 (best validation performance)

APPENDIX B: EVALUATION METRIC DEFINITIONS

For each class, let TP, FP and FN be the numbers of true positives, false positives and false negatives, and N the number of test images.

Item	Definition
Accuracy	(number of correct predictions) / N
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1-score	2 × Precision × Recall / (Precision + Recall)
Macro average	Unweighted mean of the metric over the four classes
Weighted average	Mean of the metric over the classes, weighted by the class support